

Resultant force per unit length on wire $X = (8.0 - 2.4) \times 10^{-6} = 5.6 \times 10^{-6} \text{ N m}^{-1}$ to the right

- 24 D** When the magnitude of the magnetic flux density increases, the frame experiences an increase in magnetic flux linkage out of the plane of the paper. By Lenz's law, an induced current will flow in the frame to create a magnetic field into the plane of the paper to oppose this change. Hence, this leads to a clockwise current flowing through the frame.

Option A: Sliding the rod right or left will induce an e.m.f. across the rod. Since the rod is in the centre of the metal frame, it will produce currents in opposite directions in the left and right sections of the frame.

Option B: Sliding the rod right or left will induce an e.m.f. across the rod. Since the rod is in the centre of the metal frame, it will produce currents in opposite directions in the left and right sections of the frame.

Option C: By Lenz's law, there will be an anti-clockwise induced current that produces a magnetic field out of the plane of paper to oppose the decrease in magnetic flux density.

- 25 B** Induced e.m.f. $E = B \Delta f$